

A Cluster Is Stable Across Resolutions. Does That Make It Biologically Real?

A mathematical, theoretical, and practical view of stability in single-cell RNA-seq clustering

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The central question

Suppose a group of cells remains almost unchanged when the clustering resolution is increased from 0.2 to 0.4, 0.6, 0.8, and 1.0. The branch looks persistent in a clustering tree. The cells repeatedly co-cluster. The cluster appears computationally stable.

Does that stability mean that we have discovered a biologically distinct cell population?

Not necessarily.

Abstract

Single-cell RNA sequencing can produce highly reproducible clusters, but reproducibility of a partition and biological validity of a population are different claims. A cluster can remain stable across resolution parameters, random seeds, or resampling procedures because a strong structure exists in the representation supplied to the clustering algorithm. That structure may arise from cell identity, but it may also arise from cell state, donor effects, batch, technical variation, stress responses, or a discretized segment of an underlying continuum. This article develops a mathematical and practical framework for interpreting cluster stability. We begin with graph construction and resolution-dependent community detection, distinguish several forms of stability, examine why stability is conditional on the analysis pipeline, and propose a hierarchy of evidence for moving from a computational cluster to a biological interpretation. The central principle is simple: stability is valuable evidence about robustness, but the biological meaning of the stable structure must still be established.

1 The seductive argument

Consider a typical single-cell RNA-seq analysis. At resolution 0.2 we obtain five clusters. At resolution 0.4 we obtain seven. At resolution 0.6 we obtain ten. At resolution 0.8 we obtain fourteen. At resolution 1.0 we obtain eighteen.

Most populations split as the resolution increases. Yet one group behaves differently. Nearly the same cells remain together at every resolution. In a tool such as `clustree`, the branch appears persistent and visually clean [1].

The intuitive conclusion is:

$$\text{stable cluster} \implies \text{real biological population.} \quad (1)$$

The problem is that the implication is too strong.

A more defensible statement is:

$$\boxed{\text{stable cluster} \implies \text{robust structure under the perturbations that were tested}} \quad (2)$$

The phrase *under the perturbations that were tested* is crucial. Stability across resolution values is not the same as stability across preprocessing choices, neighborhood sizes, features, donors, batches, algorithms, or experiments. Even broad computational robustness is still not identical to biological validation.

2 What does the clustering algorithm actually receive?

The clustering algorithm does not directly observe cell types. It receives a mathematical representation of cells.

Let the original gene expression matrix be

$$X \in \mathbb{R}^{n \times p}, \quad (3)$$

where n is the number of cells and p is the number of measured genes or features.

A typical analysis transforms this matrix through several steps:

$$X \longrightarrow X^* \longrightarrow Z \longrightarrow G \longrightarrow \mathcal{C}. \quad (4)$$

Here,

- X^* denotes a normalized or otherwise processed expression representation,
- $Z \in \mathbb{R}^{n \times d}$ denotes a lower-dimensional representation, often based on principal components,
- $G = (V, E, W)$ denotes a cell-cell graph,
- $\mathcal{C}$ denotes a partition of the graph into communities.

For cell i , write its lower-dimensional representation as

$$z_i = (z_{i1}, z_{i2}, \dots, z_{id}). \quad (5)$$

A distance or similarity is calculated between cells. A simple Euclidean example is

$$d_{ij} = \|z_i - z_j\|_2. \quad (6)$$

A k -nearest-neighbor graph is then constructed. In workflows such as Seurat, this may be converted to a shared-nearest-neighbor graph, and community detection is performed on that graph [2].

The conceptual pipeline is shown below.

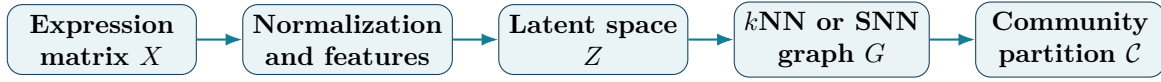


Figure 1: A cluster is the endpoint of a sequence of modeling decisions. Stability of the final partition must therefore be interpreted conditionally on the representation and graph from which it was obtained.

First principle

A clustering algorithm detects structure in the representation that it is given. It does not know whether that structure corresponds to lineage, cell state, batch, stress, treatment response, technical variation, or another source of heterogeneity.

3 What is the resolution parameter doing?

Graph-based clustering is often described operationally: lower resolution gives fewer clusters and higher resolution gives more clusters. That statement is useful, but it hides the underlying optimization problem.

For a weighted graph with adjacency matrix A , one resolution-dependent modularity objective can be written as

$$Q(\gamma) = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \gamma \frac{k_i k_j}{2m} \right] \delta(c_i, c_j), \quad (7)$$

where

$$k_i = \sum_j A_{ij}, \quad 2m = \sum_{i,j} A_{ij}, \quad (8)$$

For an undirected graph, the adjacency matrix is symmetric,

$$A_{ij} = A_{ji}.$$

Therefore, when we sum over all ordered pairs (i, j) , each undirected edge is counted twice: once as A_{ij} and once as A_{ji} . If m denotes the total number of edges, or the total edge weight in a weighted graph, counted once, then

$$\boxed{\sum_{i,j} A_{ij} = 2m} \iff \boxed{m = \frac{1}{2} \sum_{i,j} A_{ij}}.$$

A three-node triangle makes this explicit. Its adjacency matrix is

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

There are only $m = 3$ undirected edges, but

$$\sum_{i,j} A_{ij} = 6 = 2m.$$

The same fact appears in the handshaking lemma. Because

$$k_i = \sum_j A_{ij},$$

we obtain

$$\sum_i k_i = \sum_{i,j} A_{ij} = 2m.$$

Each edge contributes once to the degree, or weighted degree, of each of its two endpoints. This also explains the null-model term in modularity,

$$\frac{k_i k_j}{2m}.$$

In the configuration-model interpretation, the graph contains $2m$ edge ends, often called stubs. Node i contributes k_i of those stubs, and node j contributes k_j . Consequently, the expected connectivity between i and j is proportional to

$$\mathbb{E}[A_{ij}] \approx \frac{k_i k_j}{2m}.$$

The factor 2 is therefore not an additional correction inserted into modularity. It follows from the conventional definition of m as the total undirected edge weight counted once. If instead we define

$$M = \sum_{i,j} A_{ij} = 2m,$$

then Equation (7) can equivalently be written as

$$Q(\gamma) = \frac{1}{M} \sum_{i,j} \left[A_{ij} - \gamma \frac{k_i k_j}{M} \right] \delta(c_i, c_j).$$

The mathematics is unchanged. Only the notation has changed.

c_i denotes the community label of cell i , and

$$\delta(c_i, c_j) = \begin{cases} 1, & c_i = c_j, \\ 0, & c_i \neq c_j. \end{cases} \quad (9)$$

The parameter γ controls the scale at which communities are favored. In broad terms,

$$\gamma \uparrow \Rightarrow \text{finer partitions}, \quad (10)$$

whereas

$$\gamma \downarrow \Rightarrow \text{coarser partitions}. \quad (11)$$

This does not mean that γ is searching for a hidden integer K_{true} corresponding to the uniquely correct number of biological cell types. Rather, γ changes the scale at which the graph is partitioned.

The classical modularity framework is also known to have a resolution limit, meaning that some communities may be difficult to detect at one scale even when they are structurally meaningful [3]. This is one reason why multiresolution analysis is attractive.

A useful reframing

Instead of asking only

“What is the correct resolution?”

it is often more informative to ask

“At what biological scale am I trying to make an inference?”

4 Why multiple resolutions can all be meaningful

Biological organization is frequently hierarchical.

A coarse partition may distinguish

lymphoid from myeloid. (12)

A finer partition may distinguish

T cells, B cells, NK cells, monocytes. (13)

An even finer partition may separate

CD4 T cells and CD8 T cells, (14)

followed by naive, memory, regulatory, activated, exhausted, or other context-dependent states.

Several of these levels can be biologically useful at the same time. The scientific question determines the appropriate granularity [5].

This already tells us why a single resolution should not automatically be treated as the unique biological truth.

5 What does a clustering tree actually tell us?

Let

$$\mathcal{C}^{(\gamma)} = \{C_1^{(\gamma)}, C_2^{(\gamma)}, \dots, C_{K_\gamma}^{(\gamma)}\} \quad (15)$$

be the partition obtained at resolution γ .

A clustering tree visualizes how memberships in

$$\mathcal{C}^{(\gamma_1)}, \mathcal{C}^{(\gamma_2)}, \dots, \mathcal{C}^{(\gamma_L)} \quad (16)$$

relate to one another [1].

Importantly, the partitions are not guaranteed to form a perfectly nested hierarchy. Increasing γ can lead not only to clean splits but also to reassignments and more complicated changes in membership.

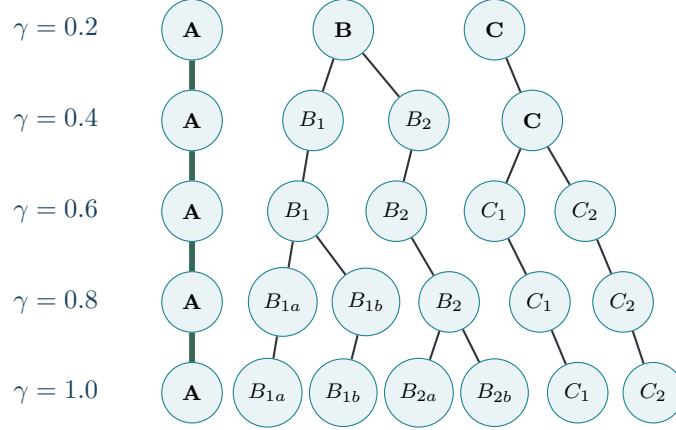


Figure 2: Conceptual clustering tree. Cluster A persists across resolutions while other communities split. Persistence is evidence of robustness to resolution, but it does not identify the biological source of that robustness.

6 Stability is not one quantity

The word *stability* is often used as if it were a single property. In fact, several distinct notions are involved.

6.1 Membership overlap across resolutions

If A and B are clusters obtained under two settings, a simple overlap measure is the Jaccard index:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}. \quad (17)$$

If $J(A, B) \approx 1$, the memberships are nearly identical. If $J(A, B) \approx 0$, they have little overlap.

However, Jaccard overlap is cluster-specific. It does not summarize agreement between two complete partitions.

6.2 Agreement between complete partitions

For two partitions U and V , one can use the Adjusted Rand Index:

$$\text{ARI}(U, V). \quad (18)$$

The ARI compares pairwise co-membership while correcting for agreement expected by chance. Values near 1 indicate strong agreement between the partitions.

Other choices include normalized mutual information or variation of information. The important point is conceptual: whole-partition similarity and single-cluster persistence are different questions.

6.3 Resampling stability

Suppose we create B perturbed or subsampled versions of the data and recluster each one. For cells i and j , define a consensus quantity

$$P_{ij} = \frac{\sum_{b=1}^B \mathbb{I}(i, j \in D^{(b)}) \mathbb{I}(c_i^{(b)} = c_j^{(b)})}{\sum_{b=1}^B \mathbb{I}(i, j \in D^{(b)})}. \quad (19)$$

Here, $D^{(b)}$ is the set of cells included in resample b .

If

$$P_{ij} \approx 1, \quad (20)$$

then cells i and j repeatedly co-cluster when they are both observed.

For a proposed cluster C , one might summarize within-cluster consensus as

$$S_{\text{within}}(C) = \frac{2}{|C|(|C| - 1)} \sum_{\substack{i < j \\ i, j \in C}} P_{ij}. \quad (21)$$

One can also examine between-cluster leakage:

$$S_{\text{between}}(C) = \frac{1}{|C|(n - |C|)} \sum_{\substack{i \in C \\ j \notin C}} P_{ij}. \quad (22)$$

An attractive pattern is

$$S_{\text{within}}(C) \gg S_{\text{between}}(C). \quad (23)$$

Even then, the conclusion is computational: the group is reproducible under that resampling scheme.

7 Why a stable cluster can still be biologically misleading

This is the central theoretical point.

Suppose the observed count for gene g in cell i is modeled as

$$Y_{ig} \sim \text{NB}(\mu_{ig}, \phi_g), \quad (24)$$

with

$$\log \mu_{ig} = \alpha_g + \beta_{g,z_i}^{\text{identity}} + s_{ig}^{\text{state}} + d_{ig}^{\text{donor}} + t_{ig}^{\text{technical}}. \quad (25)$$

The terms represent, conceptually,

- baseline expression,
- cell identity,
- cell state,
- donor or sample effects,
- technical effects.

A clustering algorithm does not observe this decomposition. It receives a transformed version of the total signal.

If state-associated variation dominates local geometry, then cells may cluster by state. If donor-associated variation dominates, they may cluster by donor. If a technical effect is strong and systematic, it may also produce a stable community.

Formally, if a source of variation H creates sufficiently strong separation in the representation Z , then the graph may satisfy

$$\mathbb{E}[W_{ij} \mid H_i = H_j] > \mathbb{E}[W_{ij} \mid H_i \neq H_j], \quad (26)$$

which encourages community detection to group cells by H .

Nothing in this inequality requires H to be a cell type.

The logical gap

A stable community demonstrates that a reproducible structure exists in the graph. It does not identify the causal or biological source of that structure.

8 Example 1: a stable interferon-response cluster

Imagine that two groups both express canonical T-cell identity genes such as

$$\text{CD3D}, \quad \text{CD3E}, \quad \text{TRAC}. \quad (27)$$

One group additionally shows strong expression of

$$\text{ISG15}, \quad \text{IFIT1}, \quad \text{IFIT3}, \quad \text{MX1}, \quad \text{OAS1}. \quad (28)$$

The interferon program can be sufficiently strong to create a dense local region in the graph. That region may persist across multiple resolutions and repeated clustering runs.

The stable object could therefore be interpreted as

$$\text{T-cell identity + interferon-responsive state,} \quad (29)$$

rather than a fundamentally new T-cell lineage.

This illustrates a key distinction:

$$\boxed{\text{cell identity} \neq \text{cell state}} \quad (30)$$

and clustering can separate either.

9 Example 2: a stable technical cluster

Suppose one library was processed separately and has a systematic shift in capture efficiency, ambient RNA contamination, mitochondrial content, or another technical property.

If that effect changes many genes coherently, it can alter the representation Z enough that cells from this library become strongly connected to one another.

The resulting cluster can be

- visually separated on UMAP,
- stable across resolution values,
- reproducible across random seeds,
- associated with highly significant differential expression.

None of those observations alone proves that it is a new biological population.

A simple contingency table of cluster membership against sample or batch can sometimes reveal the explanation immediately.

10 Stability is conditional on the perturbation

Suppose a cluster persists for

$$\gamma \in \{0.2, 0.4, 0.6, 0.8, 1.0\}. \quad (31)$$

What has actually been shown?

If every other part of the workflow is fixed, then the cluster has been tested mainly against variation in γ .

Let the full analysis be written abstractly as

$$\mathcal{C} = f(X; \theta), \quad (32)$$

where

$$\theta = (\gamma, k, d, F, N, I, A, r, \dots). \quad (33)$$

The components may represent

- γ : clustering resolution,
- k : neighborhood size,
- d : number of latent dimensions,
- F : feature set,
- N : normalization strategy,
- I : integration or batch-correction strategy,
- A : clustering algorithm or objective,
- r : random seed.

Resolution stability studies only one coordinate of this parameter space.

$$\theta = (\gamma, \theta_{-\gamma}). \quad (34)$$

Holding $\theta_{-\gamma}$ fixed while varying γ demonstrates robustness along one direction, not over the entire analysis neighborhood.

A stronger definition of computational robustness

For a proposed population C , the relevant question is not simply whether it survives a set of resolution values. It is whether an equivalent biological structure remains recognizable over a scientifically reasonable neighborhood of analysis choices:

$$\theta \in \Theta_{\text{reasonable}}.$$

11 Resolution stability versus pipeline stability

Consider a cluster that survives all tested values of γ . Now change the number of principal components from 20 to 40. The cluster disappears.

Or change

$$k = 20 \quad \text{to} \quad k = 50. \quad (35)$$

The cluster merges with a neighbor.

Or perform donor-aware integration and the cluster is no longer identifiable.

The original statement

$$\text{“Cluster 7 is stable”} \quad (36)$$

was therefore underspecified.

A more precise statement would be

“Cluster 7 is stable with respect to the tested resolution range, (37)

conditional on the chosen features, latent space, graph, and preprocessing.” (38)

Precision in language matters because different stability tests answer different questions.

12 Random-seed stability is useful, but limited

Optimization procedures can be stochastic. Repeating the clustering with seeds

$$r_1, r_2, \dots, r_R \quad (39)$$

and comparing partitions can reveal whether the optimization repeatedly converges to similar solutions.

If

$$\text{ARI}(\mathcal{C}^{(r_a)}, \mathcal{C}^{(r_b)}) \approx 1 \quad (40)$$

for many pairs of runs, the algorithmic solution is highly reproducible under seed perturbation.

This is valuable. Yet even perfect seed stability means only that the optimization behaves consistently on essentially the same graph.

It cannot establish why that graph contains the observed community structure.

13 Louvain, Leiden, and the difference between graph quality and biological meaning

Community-detection algorithms themselves also matter. The Leiden algorithm was introduced partly to address a known problem of Louvain: Louvain can return communities that are poorly connected or even disconnected. Leiden provides stronger guarantees on connectivity and local optimality under its optimization framework [4].

That mathematical improvement is important, but it produces another useful distinction:

$$\boxed{\text{well-connected graph community} \neq \text{validated biological population}} \quad (41)$$

Graph theory can tell us whether nodes form a coherent community according to a graph objective. Biology determines what that community means.

14 The hidden problem of circular validation

Suppose a cluster is discovered from the expression matrix and then differential expression is performed using the same matrix to identify genes that distinguish that cluster.

This workflow is useful for annotation, but it contains an important circular element:

$$\text{expression differences} \longrightarrow \text{cluster construction} \longrightarrow \text{expression differences between clusters.} \quad (42)$$

Strong marker significance is therefore not an independent validation of the existence of the cluster. It can describe the feature differences associated with the partition, but the strongest evidence for biological validity comes from information that is not merely a re-expression of the same clustering signal.

This is why donor replication, independent datasets, protein measurements, spatial context, perturbation, lineage information, or functional assays can add qualitatively different evidence.

15 A stable cluster should be checked across donors

Suppose Cluster 8 contains 1,200 cells. That sounds substantial.

Now inspect the sample composition.

Donor	Scenario A	Scenario B
D1	204	1,176
D2	186	4
D3	221	6
D4	193	3
D5	207	7
D6	189	4

Both scenarios contain 1,200 cells.

In Scenario A, the population is represented across six donors. In Scenario B, it is almost entirely donor-specific.

These situations support different claims.

A donor-specific population may still be biologically meaningful. For example, a tumor-specific subclone, infection-specific response, or patient-specific genotype effect can be scientifically important. The key issue is that the biological claim must match the replication structure.

If the claim is

$$\text{“this is a general human cell type,”} \quad (43)$$

replication across independent individuals matters greatly.

If the claim is

“this is a patient-specific malignant subpopulation,” (44)

single-donor restriction is not automatically disqualifying.

Replication must match the claim

There is no universal rule that every biologically meaningful cluster must appear in every donor. Instead, ask whether the observed sample distribution is compatible with the biological statement you want to make.

16 Marker coherence matters more than a spectacular p value

Suppose a cluster contains hundreds of significantly differentially expressed genes, with some adjusted p values smaller than

$$10^{-30}. \quad (45)$$

This may look decisive, but statistical significance alone does not establish a cell identity.

A stronger biological interpretation comes from a coherent program. For example,

$$g_1 \uparrow, \quad g_2 \uparrow, \quad g_3 \uparrow, \quad g_4 \uparrow \quad (46)$$

may jointly support a known lineage, pathway, functional state, or regulatory module.

The question is not merely

$$\text{“Which genes are significant?”} \quad (47)$$

but rather

$$\boxed{\text{“What coherent biological process explains the distinguishing signal?”}} \quad (48)$$

This distinction is particularly important in single-cell data, where very large cell counts can produce extremely small cell-level p values even when the biological replication is modest.

17 Discrete clusters can emerge from continuous biology

Now consider a deeper problem.

Suppose the true biological system is a continuous trajectory parameterized by

$$t \in [0, 1], \quad (49)$$

with expression

$$X = f(t) + \varepsilon. \quad (50)$$

There may be no true biological discontinuity at $t = 0.35$ or $t = 0.62$.

Nevertheless, graph community detection is asked to produce discrete groups. It may partition the continuum as

$$C_1, C_2, C_3, C_4. \quad (51)$$

These partitions may even be stable because local graph connectivity changes gradually and consistently along the manifold.

Therefore,

$$\boxed{\text{stable discrete partition} \not\Rightarrow \text{discrete biological generative states}} \quad (52)$$

This matters in differentiation, development, activation, disease progression, and other processes where cells naturally occupy continua.

18 A generative thought experiment

Compare two hypothetical datasets.

18.1 Dataset A: genuinely separated populations

Suppose

$$X \sim \sum_{k=1}^3 \pi_k \mathcal{N}(\mu_k, \Sigma_k), \quad (53)$$

with well-separated means μ_1, μ_2, μ_3 .

Here, robust neighborhoods and stable partitions are expected because the data are generated from discrete components.

18.2 Dataset B: a continuous manifold

Suppose instead

$$X = f(t) + \varepsilon, \quad t \in [0, 1]. \quad (54)$$

A graph algorithm can still divide the manifold into several communities.

Thus, the observation

“the graph contains stable communities” (55)

does not uniquely identify the generating model.

The same computational phenomenon can be compatible with

discrete populations (56)

or

a discretized continuum. (57)

This is an identifiability problem: different biological mechanisms can produce similar clustering behavior.

19 Why instability is not automatically failure

The reverse mistake is also common.

If a cluster is unstable, it is tempting to conclude that the cluster is meaningless.

But instability can arise because

- the population is rare,
- the population lies between two larger states,
- the biological process is continuous,
- the distinguishing genes are weakly sampled,
- dropout or low counts reduce neighborhood consistency,
- the chosen representation is not optimized for the relevant biology.

A rare but experimentally validated cell type may be less stable under subsampling than a large batch-driven cluster.

Therefore,

stability is neither necessary nor sufficient for biological reality (58)

in the strict logical sense.

It is evidence, not a verdict.

20 Do not use UMAP separation as independent proof

A common argument is:

“The cluster is clearly separated on UMAP, so it is real.” (59)

This is weak reasoning.

UMAP is a nonlinear visualization of a representation that is itself related to the neighborhood structure used during analysis. Visual separation is therefore not an independent biological experiment.

A separated island can be informative, but it should be treated as descriptive evidence, not as proof of cell identity.

The relevant question remains: what underlying signal produced the neighborhood structure, and does independent evidence support its biological interpretation?

21 A hierarchy of evidence

A useful way to think about cluster interpretation is to separate computational evidence from biological evidence.

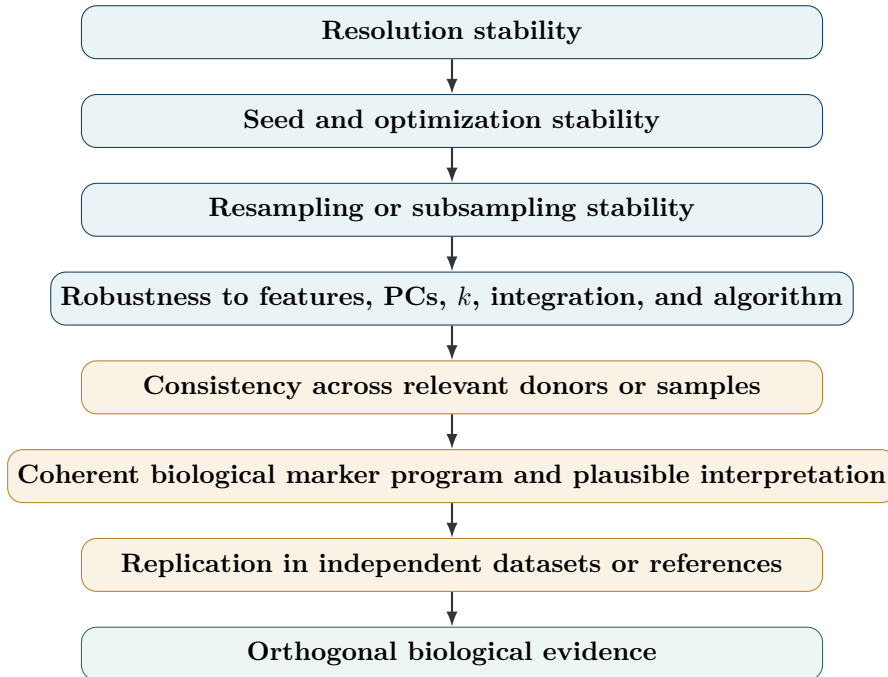


Figure 3: A conceptual evidence ladder. Evidence becomes progressively less dependent on the original clustering construction as we move toward independent biological validation.

The purpose of this hierarchy is not to create a universal checklist in which every study must reach the final step. Rather, it helps match the strength of the claim to the strength and independence of the evidence.

22 A multidimensional robustness profile is better than one stability score

Because stability is multidimensional, compressing everything into one number can hide important weaknesses.

For a proposed cluster C , define a robustness profile

$$\mathbf{R}(C) = (S_\gamma, S_{\text{seed}}, S_{\text{resample}}, S_{\text{parameter}}, S_{\text{sample}}, S_{\text{external}}). \quad (60)$$

Here,

- S_γ measures robustness across resolution values,
- S_{seed} measures robustness across optimization seeds,
- S_{resample} measures robustness under cell or sample perturbation,
- $S_{\text{parameter}}$ measures sensitivity to reasonable pipeline choices,
- S_{sample} summarizes reproduction across relevant biological samples,
- S_{external} summarizes replication outside the discovery dataset.

The exact numerical definitions can vary with the study. The conceptual advantage is that a profile reveals *where* the evidence is strong and where it is weak.

For example,

$$\mathbf{R}(C) = (\text{high}, \text{high}, \text{high}, \text{low}, \text{low}, \text{unknown}) \quad (61)$$

communicates something very different from

$$\mathbf{R}(C) = (\text{high}, \text{high}, \text{high}, \text{high}, \text{high}, \text{high}). \quad (62)$$

A single stability number would conceal that distinction.

23 What should we test in practice?

For an interesting cluster C^* , I would separate the analysis into six questions.

23.1 Question 1: Does the cluster persist across granularity?

Examine a sensible range of clustering resolutions. Use a clustering tree to determine whether the population persists, splits gradually, merges abruptly, or exchanges substantial membership with neighboring communities.

Do not focus only on the number of clusters. Focus on the lineage of the specific population of interest across parameter values.

23.2 Question 2: Does the cluster survive perturbation of the cells?

Subsample cells or bootstrap appropriate units and recluster. Examine cluster-level Jaccard similarity or consensus co-clustering probabilities.

For rare populations, interpret instability carefully because subsampling can remove much of the evidence supporting the group.

23.3 Question 3: Does the biological conclusion survive reasonable pipeline choices?

Vary parameters that could materially affect graph geometry, such as

$$\text{number of PCs, } k, \text{ feature set, integration strategy.} \quad (63)$$

The goal is not to demand identical cell labels under every setting. The goal is to determine whether the proposed biological structure remains recognizable.

23.4 Question 4: Is the cluster replicated at the sample level?

Inspect

$$P(\text{sample} \mid C^*). \quad (64)$$

Ask whether the sample distribution supports the intended biological claim.

If the cluster is disease-specific, ask whether it appears across multiple disease samples. If it is claimed as a general cell type, ask whether it appears across multiple independent individuals or datasets.

23.5 Question 5: What biological program defines the cluster?

Do the distinguishing genes form a coherent lineage, pathway, regulatory, functional, or state-specific program?

Ask whether the signal is better explained by

$$\text{identity, state, trajectory, artifact.} \quad (65)$$

This classification is often more useful than immediately forcing every cluster into a cell-type label.

23.6 Question 6: Is there independent support?

Possible orthogonal evidence includes

- protein measurements such as CITE-seq or flow cytometry,
- spatial localization,
- histological context,
- lineage tracing,
- perturbation experiments,
- functional assays,

- replication in an independent cohort or technology.

The more independent the evidence is from the original clustering matrix, the more strongly it can support a biological interpretation.

24 A practical decision framework

The following framework is intentionally conservative but not rigid.

Observation	What it supports	What it does not establish
Stable across resolution	Robustness to clustering granularity	Cell-type identity
Stable across random seeds	Optimization reproducibility	Biological replication
Stable under resampling	Membership robustness to data perturbation	Biological mechanism
Stable across pipeline choices	Broader computational robustness	Independent biological validation
Present across relevant donors	Sample-level reproducibility	Functional identity
Coherent marker program	Biological plausibility	Causal distinctness
Replicated externally	Generalizability beyond discovery data	Mechanistic necessity
Orthogonal assay support	Independent biological evidence	Universal truth across all contexts

25 How strong should the language be?

Scientific language should scale with the evidence.

If the evidence is mainly computational, a careful description might be

We identified a reproducible transcriptional cluster that persisted across a range of graph-clustering resolutions.

If the population is additionally replicated across donors and supported by a coherent program, one might say

We identified a reproducible transcriptionally distinct population consistent with a specific cellular state or subtype.

A claim such as

We discovered a novel cell type.

should generally require substantially stronger biological evidence than resolution stability alone.

The issue is not linguistic caution for its own sake. It is alignment between the strength of the claim and the independence of the evidence.

26 When stability analysis is genuinely valuable

None of this means that stability analysis is unimportant. Quite the opposite.

If a cluster disappears whenever

$$\gamma : 0.5 \longrightarrow 0.6, \quad (66)$$

or changes radically under mild subsampling, that sensitivity deserves attention.

Clustering trees are useful because they expose how partitions evolve across granularity rather than hiding that information behind a single chosen resolution [1]. Seed perturbation can reveal optimization uncertainty. Resampling can reveal weak membership. Parameter perturbation can expose dependence on arbitrary analysis choices.

The error is not in asking whether the cluster is stable.

The error is asking stability to answer a biological question that it was not designed to answer.

27 A deeper statistical formulation

We can formalize the distinction between computational and biological questions.

Let C denote the event that a recognizable cluster is recovered after an analysis perturbation Π . Computational stability concerns a quantity of the form

$$P(C \mid X, \Pi, \mathcal{M}), \quad (67)$$

where $\mathcal{M}$ denotes the analysis framework.

Biological validity concerns a different latent proposition, say B , such as

$$B = \text{ext} \text{“these cells constitute a reproducible biological population relevant to the scientific claim.”} \quad (68)$$

The desired inference is about

$$P(B \mid \text{all available evidence}). \quad (69)$$

High computational stability can increase confidence in B , but it is only one component of the evidence. In particular,

$$P(C \mid X, \Pi, \mathcal{M}) \approx 1 \quad (70)$$

does not imply

$$P(B \mid X) \approx 1. \quad (71)$$

The distinction is elementary in probability but easy to lose in practice because the visual output of clustering appears categorical and biologically suggestive.

28 The most important question is not “Is the cluster stable?”

A stronger sequence of questions is:

1. What signal in the data created this partition?
2. Is that signal robust to reasonable computational perturbations?
3. Does the structure reproduce at the biological sample level appropriate to the claim?
4. Is the signal better explained by identity, state, trajectory, technical structure, or another source?
5. Does independent evidence support the interpretation?

This changes the analytical logic from

$$\text{cluster} \longrightarrow \text{marker} \longrightarrow \text{name} \quad (72)$$

into

$$\boxed{\text{structure} \longrightarrow \text{robustness} \longrightarrow \text{biological explanation} \longrightarrow \text{validation}} \quad (73)$$

That is a much stronger scientific workflow.

29 Conclusion

A cluster that persists across multiple resolutions is worth paying attention to. It tells us that the group is not simply an arbitrary artifact of one narrow value of the resolution parameter.

But resolution stability tells us something specific:

The grouping persists when the granularity of a particular graph-clustering analysis is changed.

It does not, by itself, tell us why the cells group together.

A genuine cell type can be stable.

A treatment-induced state can be stable.

A batch effect can be stable.

A cell-cycle program can be stable.

A stress response can be stable.

A discretized segment of a continuous trajectory can also be stable.

The computational task is therefore not merely to find the most stable partition. The scientific task is to identify what generates that stability and whether the source of the structure is biologically meaningful for the claim being made.

Take-home message

Do not ask only: “Is this cluster stable?”

Ask instead:

“What is stable about these cells, under which perturbations, and is the source of that stability biologically meaningful?”

A short LinkedIn version

Your scRNA-seq cluster survives resolution 0.2, 0.4, 0.6, 0.8, and 1.0.

Does that mean you have found a biologically real population?

Not necessarily.

Cluster stability tells us that a computational structure is reproducible under the perturbation we tested. It does not tell us what generated that structure.

A cell type can be stable. But so can a batch effect, cell-cycle program, stress response, interferon response, or treatment-induced state.

And a cluster that is stable across resolution can still disappear when we change the neighborhood size, feature set, number of PCs, integration strategy, or biological samples.

So the deeper question is not only

“Is this cluster stable?”

but

“What is generating the stability?”

A detailed look at the mathematics of graph clustering, resolution, clustering trees, resampling stability, donor reproducibility, continuous biological manifolds, and the evidence needed before moving from a *stable cluster* to a *biological population*.

References

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